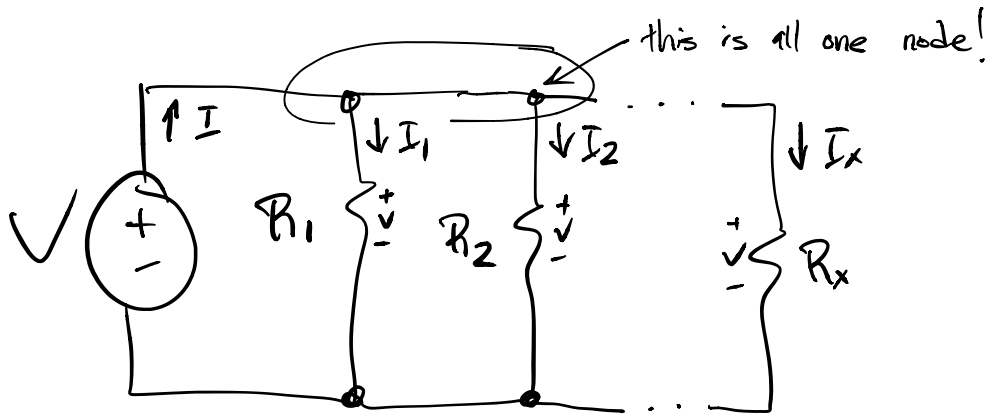


# Resistors in Parallel

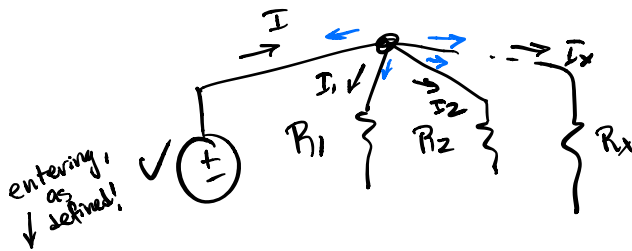
Consider a number of resistors in parallel, driven from a voltage source:



each resistor draws a current by Ohm's Law:

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_x = \frac{V}{R_x}$$

by Kirchhoff's Current Law (top node)



$$-I + I_1 + I_2 + \dots + I_x = 0$$

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \dots + \frac{V}{R_x}$$

$$I = V \left[ \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_x} \right]$$

.. the equivalent resistance "seen" by the voltage source is  $R = \frac{V}{I}$

$$\therefore R_{\text{parallel}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_x}}$$

alt. way of looking at it:

.. conductances in parallel add!

$$G_{\text{parallel}} = G_1 + G_2 + \dots + G_x$$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_x}$$

$$\therefore R_{\text{parallel}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_x}}$$

~ Ultra - useful simplification that only works for two resistors:

$$R_{||} = \frac{R_1 R_2}{R_1 + R_2}$$

product-over-sum

↑  
alt. notation  
for parallel!

.. if equal-value resistors:  $\left( \frac{R}{n} \right)$

↓

# System of Preferred Values

- .. used in a wide variety of products and processes!
- .. international standards (ISO, etc..) on everything from electronic components to fasteners, etc..

## E6 series of component values

- .. corresponds to a 20% tolerance of value
- .. every decade (factor of ten) has this set of values:

10

15

22

33

47

68

∴ a 20% tolerance resistor is available in only these values, any order of magnitude

$\infty$  100 $\Omega$  150 $\Omega$  220 $\Omega$  330 $\Omega$  etc..  
 1k $\Omega$  1.5k $\Omega$  2.2k $\Omega$  ...

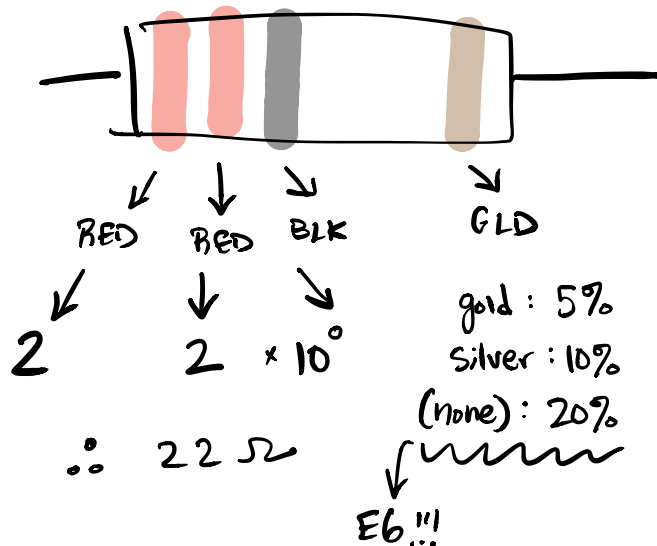
220k 330k 470k

.. logarithmic progression of values, just overlap within tolerances

a 10 $\Omega$  resistor  $\oplus$  20%  $\rightarrow$  12 $\Omega$

a 15 $\Omega$  resistor  $\ominus$  20%  $\rightarrow$  12 $\Omega$  just overlap

.. nobody makes 20% resistors anymore, but some components like electrolytic capacitors still use E6 values because their tolerance is often  $\pm 20\%$ !



### color codes

|   |     |       |
|---|-----|-------|
| 0 | BLK | bad   |
| 1 | BRN | beer  |
| 2 | RED | rots  |
| 3 | ORG | our   |
| 4 | YEL | young |
| 5 | GRN | guts  |
| 6 | BLU | but   |
| 7 | VIO | vodka |
| 8 | GRY | goes  |
| 9 | WHT | well  |

↑  
official abbreviations?

## E12 Values $\rightarrow$ 10% tolerance (resistors: silver tolerance band)

10  
12  
15  
18  
22  
27  
33  
39  
47  
56  
68  
82

ex: 100k $\Omega$  resistor @ +10%

$\rightarrow$  110k

120k $\Omega$  resistor @ -10%

$\rightarrow$  108k

(good enough @ two sig. dig.s!)

.. Film capacitors use E12 values etc..

## E24 values $\rightarrow$ 5% tolerance

|    |    |    |
|----|----|----|
| 10 | 24 | 56 |
| 11 | 27 | 62 |
| 12 | 30 | 68 |
| 13 | 33 | 75 |
| 15 | 36 | 82 |
| 16 | 39 | 91 |
| 18 | 43 |    |
| 20 | 47 |    |
| 22 | 51 |    |

gold-band carbon film resistors in hb!

.. intermediate values between E12

.. if you need higher precision than 5% :

E48  $\rightarrow$  2% values

E96  $\rightarrow$  1% values

$\rightarrow$  precision metal film resistors

.. I don't have all these memorized!

table (on Pilot)

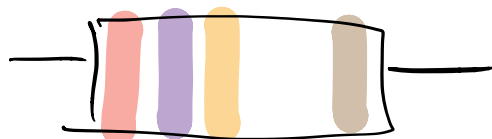
three significant digits

.. Very important to specify components in your designs that are actually available!!

ex: you compute 26.942 k $\Omega$

nearest E24 value, assuming 5% resistor is acceptable :

27 k $\Omega$



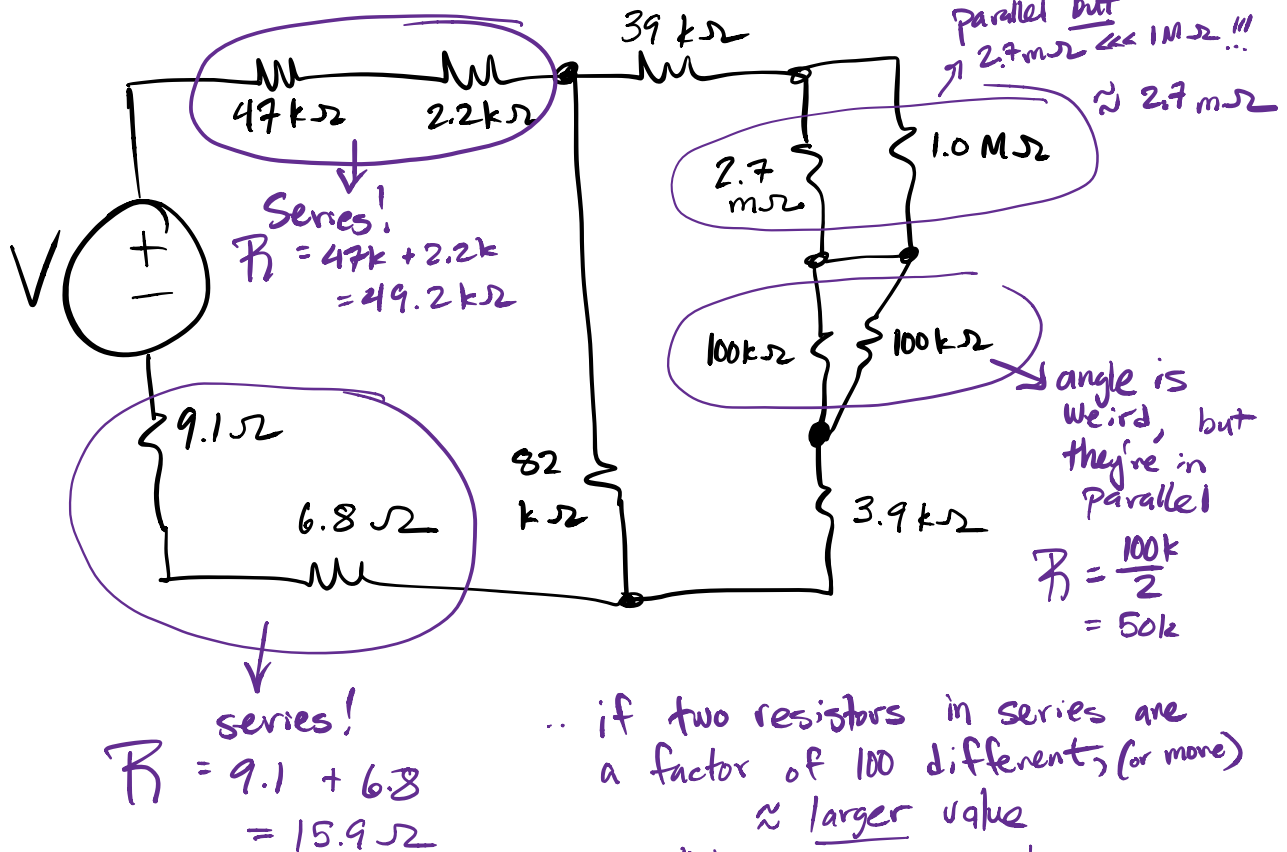
27  $\times 10^3$  ( $\Omega$ )

RES 110 086

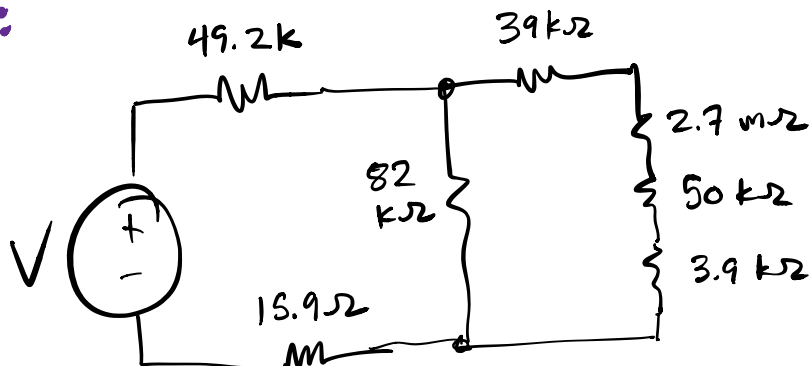
E192  
0.5%

# Resistor Networks

.. randomly pick E24 resistors, any order of magnitude



redraw:



homework: Continue this process to determine the equivalent resistance seen by V.